

DD_SIP-03 - Discount Assignment

Version: 1.0 Status: Draft for review Bundle: 09_catalogs Audience: commercial developers, billing developers, catalog developers, frontend developers, QA Related DDs:

- DD_PLM-CFG-04 - Discount Catalog
- DD_SIP-01 - Package Management
- DD_SIP-05 - Promotional Campaigns
- DD_DIS-OP-01 - Discount Runtime Application
- DD_BIL-01 - Charging Engine Core
- DD_BIL-02 - Invoicing Core
- DD_BIL-05 - Wallet and Topup
- DD_EM-01 - Franchise Management
- DD_SALES-01 - Lead, Territory, and Franchise Sales Management
- DD_EM-CFG-04 - Approval Workflow Catalog
- FE-APP-01 - BSS Backoffice Web App
- FE-APP-02 - Outdoor Sales Franchise App

MVP simplification baseline: first release supports direct discount assignment/cancellation with approval where required. Bulk assignment, complex campaign chaining, and advanced exception workflows are phased after the core billing path is stable.

1. Purpose

SIP-03 defines how SOPHIX assigns configured discounts to customers, accounts, subscriptions, orders, franchises, or campaign cohorts.

The discount catalog defines what a discount is. SIP-03 defines who gets it, why they get it, for which scope, for which period, and how the assignment is cancelled or audited.

Examples:

- retention agent assigns 10% recurring discount to a subscription for 3 months;
- sales campaign assigns free installation discount to new customers in a territory;
- franchise manager assigns acquisition discount to leads converted by a franchise;
- care supervisor assigns goodwill discount after approved complaint;
- batch job assigns a campaign discount to eligible customers;
- admin cancels a customer discount assignment because customer no longer qualifies.

SIP-03 does not calculate invoice amounts at billing time. Billing-time application is owned by DIS-OP-01 and BIL-01/BIL-02. SIP-03 creates and governs assignment records that runtime billing reads.

2. Ownership Boundary

Concern	Owner
Discount definition, amount/type/flavor/eligibility catalog	PLM-CFG-04
Discount assignment lifecycle and audit	SIP-03
Campaign eligibility and campaign-owned mass assignment	SIP-05
Billing-time discount calculation/application	DIS-OP-01 / BIL-01
Invoice generation using applied discount lines	BIL-02
Wallet credit posting for wallet-credit discounts	BIL-05 / DIS-OP-01
Franchise/agent eligibility	EM-01 / SALES-01

Concern	Owner
Configurable approval process, approval tasks, approval callbacks	EM-CFG-04
Backoffice UI	FE-APP-01
Sales app discount display/request	FE-APP-02

Rule: SIP-03 may create, approve, activate, cancel, and expire discount assignments. It must not apply money directly to invoices or wallets. Runtime financial impact is handled by billing modules.

Approval rule: SIP-03 does not define approval steps locally. It calls EM-CFG-04 to evaluate and run the configured approval policy for actions such as DISCOUNT_ASSIGNMENT_CREATE, DISCOUNT_ASSIGNMENT_CANCEL, or DISCOUNT_ASSIGNMENT_OVERRIDE. EM-CFG-04 may execute the approval as auto-approval, single-step DB-driven approval, multi-step approval, or Camunda BPMN depending on operator policy.

3. Key Concepts

Concept	Meaning
Discount catalog item	Configured discount definition from PLM-CFG-04.
Assignment	A record granting a discount to a target customer/account/subscription/order/franchise/cohort.
Scope	What the assignment applies to: customer, account, subscription, order, package, franchise, campaign cohort.
Validity window	Dates during which assignment can be used by runtime.
Assignment reason	Business reason: acquisition, retention, complaint compensation, staff, franchise offer, migration, manual goodwill.
Campaign binding	Link between assignment and campaign. Campaign owns eligibility; SIP-03 owns assignment record.
Runtime application	The billing event where an active assignment actually reduces a charge or posts a wallet credit. Owned by DIS-OP-01.

4. High Level Flow

```

flowchart TD
    A["User/system requests discount assignment"] --> B["Load discount catalog item"]
    B --> C["Validate target and eligibility"]
    C --> D{"Approval required?"}
    D --> E["Create approval task"]
    E --> F["Create assignment DRAFT/PENDING_APPROVAL"]
    F --> G["Activate assignment"]
    G --> H["Publish DiscountAssigned event"]
    H --> I["DIS-OP/BIL runtime reads active assignment"]
    I --> J["Apply discount during charge/invoice/topup event"]
    J --> K["Assignment expires or is cancelled"]
    K --> L["Publish status event"]

    A --> B --> C --> D
    D -->|Yes| E --> F
    D -->|No| F
    F --> G --> H --> I --> J
    G --> K --> L

```

5. Assignment Scopes

Scope	Use	Runtime behavior
CUSTOMER	Discount can apply to any eligible service owned by customer.	Runtime selects matching subscriptions/events.
ACCOUNT	Discount applies to account-level charges.	Runtime checks account id.
SUBSCRIPTION	Discount applies to one subscription.	Most common recurring discount.
ORDER	Discount applies during onboarding/order charges.	Runtime checks order/pro forma/first invoice.
PACKAGE	Discount applies when target package is active.	Runtime checks package ref.

Scope	Use	Runtime behavior
FRANCHISE	Discount applies to customers/leads from a franchise.	Used for acquisition campaigns.
CAMPAIGN_COHORT	Discount applies to members of a campaign cohort.	Runtime checks assignment generated by campaign.

6. Tables

6.1 discount_assignment

Stores the assignment lifecycle.

Use this table whenever a discount is granted to a target. Runtime billing reads only active/effective rows through APIs or read models.

Column	Type	Purpose	Sample
assignment_id	UUID PK	Assignment id.	da-001
operator_code	TEXT	Operator/tenant.	WIK
discount_code	TEXT	Discount catalog code from PLM-CFG-04.	RETENTION_10PCT_3M
discount_id	TEXT	Discount catalog id/version ref.	disc-1001
scope_type	TEXT	CUSTOMER, ACCOUNT, SUBSCRIPTION, ORDER, PACKAGE, FRANCHISE, CAMPAIGN_COHORT.	SUBSCRIPTION
scope_ref_id	TEXT	Id of the scoped target.	SUB-700045
customer_id	TEXT NULL	Customer id when known.	CUS-100045
account_id	TEXT NULL	Account id when known.	ACC-900045
subscription_id	TEXT NULL	Subscription id when known.	SUB-700045
package_ref	TEXT NULL	Package ref when package-scoped.	pkg-fiber-100m
campaign_id	TEXT NULL	Campaign that created assignment.	camp-retention-june
franchise_id	TEXT NULL	Franchise attribution.	fr-019
reason_code	TEXT	Business reason.	RETENTION_SAVE
source_channel	TEXT	BACKOFFICE, SALES_APP, CAMPAIGN, ASR, BATCH, API.	BACKOFFICE
valid_from	DATE	Assignment start date.	2026-06-01
valid_to	DATE NULL	Assignment end date.	2026-08-31
status	TEXT	DRAFT, PENDING_APPROVAL, ACTIVE, SUSPENDED, EXPIRED, CANCELLED, REJECTED.	ACTIVE
priority	INT	Runtime ordering when multiple discounts match.	100
stacking_group_code	TEXT NULL	Group used to prevent incompatible stacking.	RECURRING_MRC
metadata_json	JSONB	Extra source context.	{"approvedBy": "usr-supervisor-002"}
created_by_user_id	TEXT	Creator user/system.	usr-care-001
created_at	TIMESTAMP	Creation timestamp.	2026-06-01T09:00:00Z
activated_at	TIMESTAMP NULL	Activation timestamp.	2026-06-01T09:05:00Z
cancelled_at	TIMESTAMP NULL	Cancellation timestamp.	null

Recommended indexes:

- (operator_code, status, valid_from, valid_to).
- (operator_code, scope_type, scope_ref_id, status).
- (customer_id, status).
- (subscription_id, status).

- (campaign_id, status).

6.2 discount_assignment_approval

Stores the SIP-03 side reference to a shared EM-CFG-04 approval request.

Use this table when operator policy requires approval for high-value, long-duration, manual, or complaint-related discounts. The configurable policy, approval steps, approver rules, BPMN process key, user tasks, and final decision are owned by EM-CFG-04. This table lets SIP-03 block activation until the shared approval request returns an approved outcome.

Column	Type	Purpose	Sample
approval_id	UUID PK	Approval id.	daa-001
assignment_id	UUID	Assignment id.	da-001
approval_policy_code	TEXT	EM-CFG-04 policy code used.	DISCOUNT_ASSIGNMENT_RETENTION
approval_request_id	TEXT NULL	EM-CFG-04 runtime approval request id.	apr-da-001
approval_mode	TEXT	EM-CFG-04 mode: AUTO, SINGLE_STEP, MULTI_STEP, BPMN.	BPMN
camunda_process_instance_id	TEXT NULL	Camunda instance id when EM-CFG-04 uses BPMN.	cam-proc-da-001
status	TEXT	PENDING, APPROVED, REJECTED, CANCELLED.	APPROVED
assigned_queue_code	TEXT	Approval queue.	CARE_SUPERVISOR
actor_user_id	TEXT NULL	Approver.	usr-supervisor-002
decision_comment	TEXT NULL	Decision note.	Approved 3 month retention save.
created_at	TIMESTAMP	Creation timestamp.	2026-06-01T09:01:00Z
decided_at	TIMESTAMP NULL	Decision timestamp.	2026-06-01T09:04:00Z

Recommended indexes:

- (assignment_id, status).
- (approval_request_id).
- (assigned_queue_code, status).

6.3 discount_assignment_batch

Stores bulk assignment jobs.

Use this table for campaign, CSV upload, migration, or admin bulk operations.

Column	Type	Purpose	Sample
batch_id	UUID PK	Batch id.	dab-001
operator_code	TEXT	Operator/tenant.	WIK
batch_type	TEXT	CSV_UPLOAD, CAMPAIGN, SEGMENT, MIGRATION, API.	CAMPAIGN
discount_code	TEXT	Discount catalog code.	ACQ_FREE_INSTALL
campaign_id	TEXT NULL	Campaign source.	camp-acq-june
requested_by_user_id	TEXT	User/system.	usr-commercial-001
status	TEXT	DRAFT, VALIDATING, READY, RUNNING, COMPLETED, PARTIALLY_FAILED, FAILED, CANCELLED.	RUNNING
total_rows	INT	Total input targets.	1200
valid_rows	INT	Rows passing validation.	1182
failed_rows	INT	Rows failed.	18
created_at	TIMESTAMP	Creation timestamp.	2026-06-01T08:00:00Z
completed_at	TIMESTAMP NULL	Completion timestamp.	2026-06-01T08:20:00Z

Recommended indexes:

- (operator_code, status, created_at DESC).
- (campaign_id).

6.4 discount_assignment_batch_item

Stores per-target result for a bulk assignment batch.

Use this table so failed rows can be reviewed and retried without reprocessing successful rows.

Column	Type	Purpose	Sample
batch_item_id	UUID PK	Batch item id.	dabi-001
batch_id	UUID	Batch id.	dab-001
target_ref_type	TEXT	CUSTOMER, ACCOUNT, SUBSCRIPTION, ORDER, FRANCHISE.	SUBSCRIPTION
target_ref_id	TEXT	Target id.	SUB-700045
status	TEXT	PENDING, VALID, CREATED, FAILED, SKIPPED.	CREATED
assignment_id	UUID NULL	Created assignment.	da-001
failure_code	TEXT NULL	Failure reason.	DISCOUNT_NOT_ELIGIBLE_FOR_PACKAGE
failure_detail	TEXT NULL	Human-readable failure.	Target package is not eligible.

Recommended indexes:

- (batch_id, status).
- (target_ref_type, target_ref_id).

6.5 discount_assignment_status_history

Stores immutable assignment status changes.

Use this table for audit, reporting, and dispute investigation.

Column	Type	Purpose	Sample
history_id	UUID PK	History id.	dash-001
assignment_id	UUID	Assignment id.	da-001
old_status	TEXT NULL	Previous status.	PENDING_APPROVAL
new_status	TEXT	New status.	ACTIVE
reason_code	TEXT	Reason for transition.	APPROVED
changed_by_user_id	TEXT	Actor/system.	usr-supervisor-002
changed_at	TIMESTAMP	Change timestamp.	2026-06-01T09:05:00Z

Recommended indexes:

- (assignment_id, changed_at DESC).

7. APIs

7.1 Create Assignment

```
POST /api/discount-assignments
Idempotency-Key: da-retention-sub-700045-20260601
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "discountCode": "RETENTION_10PCT_3M",
  "scopeType": "SUBSCRIPTION",
  "scopeRefId": "SUB-700045",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
```

```

    "subscriptionId": "SUB-700045",
    "reasonCode": "RETENTION_SAVE",
    "sourceChannel": "BACKOFFICE",
    "validFrom": "2026-06-01",
    "validTo": "2026-08-31",
    "priority": 100,
    "stackingGroupCode": "RECURRING_MRC"
  }

```

Response:

```

{
  "assignmentId": "da-001",
  "status": "ACTIVE",
  "approvalRequired": false
}

```

7.2 Preview Assignment

```

POST /api/discount-assignments/preview
Content-Type: application/json

```

Returns whether the target is eligible and whether approval is required. SIP-03 determines this by calling EM-CFG-04 approval evaluation with a business context such as discount code, duration, reason code, source channel, customer segment, franchise, estimated value, and requester role.

Example EM-CFG-04 evaluation call:

```

POST /api/approvals/evaluate
Content-Type: application/json

{
  "operatorCode": "WIK",
  "moduleCode": "SIP-03",
  "actionCode": "DISCOUNT_ASSIGNMENT_CREATE",
  "subjectType": "DISCOUNT_ASSIGNMENT",
  "context": {
    "discountCode": "RETENTION_10PCT_3M",
    "reasonCode": "RETENTION_SAVE",
    "sourceChannel": "BACKOFFICE",
    "scopeType": "SUBSCRIPTION",
    "durationDays": 92,
    "estimatedValue": 1500.00,
    "currencyCode": "KES"
  }
}

```

If EM-CFG-04 returns `approvalRequired = true`, SIP-03 creates the assignment in `PENDING_APPROVAL`, starts an approval request through EM-CFG-04, and stores `approval_request_id`.

7.3 Approve Assignment

Normal approver action happens through EM-CFG-04 task APIs:

```

POST /api/approvals/tasks/{approval_task_id}/approve
Content-Type: application/json

{
  "decisionComment": "Approved retention discount for three months."
}

```

After EM-CFG-04 finalizes the approval request, it calls the SIP-03 callback:

```

POST /internal/discount-assignments/{assignment_id}/approval-outcome
Content-Type: application/json

{
  "approvalRequestId": "apr-da-001",
  "outcome": "APPROVED",
  "decisionComment": "Approved retention discount for three months."
}

```

SIP-03 then marks `discount_assignment_approval.status = APPROVED` and activates the assignment. If outcome is `REJECTED`, SIP-03 marks the assignment `REJECTED`.

7.4 Cancel Assignment

```

POST /api/discount-assignments/{assignment_id}/cancel
Content-Type: application/json

{

```

```

    "reasonCode": "CUSTOMER_NO_LONGER_ELIGIBLE",
    "effectiveDate": "2026-07-01",
    "comment": "Customer migrated to package not eligible for this discount."
  }
}

```

Cancellation stops future runtime application. It does not reverse already-applied invoice/wallet impact. Reversal requires BIL adjustment.

7.5 Bulk Assignment

```

POST /api/discount-assignment-batches
Content-Type: application/json

{
  "operatorCode": "WIK",
  "batchType": "CAMPAIGN",
  "discountCode": "ACQ_FREE_INSTALL",
  "campaignId": "camp-acq-june",
  "validFrom": "2026-06-01",
  "validTo": "2026-06-30",
  "targets": [
    {"targetRefType": "ORDER", "targetRefId": "ORD-1001"},
    {"targetRefType": "ORDER", "targetRefId": "ORD-1002"}
  ]
}

```

7.6 Runtime Query API

Runtime billing modules query active assignments:

```
GET /api/discount-assignments/effective?operatorCode=WIK&subscriptionId=SUB-700045&eventCode=RECURRING_MRC&onDate=2026-06-01
```

Response:

```

{
  "assignments": [
    {
      "assignmentId": "da-001",
      "discountCode": "RETENTION_10PCT_3M",
      "scopeType": "SUBSCRIPTION",
      "scopeRefId": "SUB-700045",
      "priority": 100,
      "stackingGroupCode": "RECURRING_MRC",
      "validFrom": "2026-06-01",
      "validTo": "2026-08-31"
    }
  ]
}

```

DIS-OP-01 uses this result to calculate application. SIP-03 does not calculate money.

8. Events

Event	Topic	When emitted
DiscountAssignmentCreated	sophix.commercial.discount-assignment.created	Assignment created.
DiscountAssignmentApprovalRequired	sophix.commercial.discount-assignment.approvalrequired	Approval opened.
DiscountAssignmentActivated	sophix.commercial.discount-assignment.activated	Assignment becomes active.
DiscountAssignmentCancelled	sophix.commercial.discount-assignment.cancelled	Assignment cancelled.
DiscountAssignmentExpired	sophix.commercial.discount-assignment.expired	Assignment passes valid_to.
DiscountAssignmentBatchCompleted	sophix.commercial.discount-assignment.batchcompleted	Batch finishes.

9. Business Rules

Rule	Description	Reason
R-SIP-DA-01	Discount code must exist and be active in PLM-CFG-04.	Prevent assigning retired discounts.
R-SIP-DA-02	Target scope reference must exist in owning module.	Prevent orphan assignments.
R-SIP-DA-03	Target must satisfy discount catalog eligibility: package, franchise, billable event, wallet, geography.	Commercial control.
R-SIP-DA-04	Assignment validity must be inside discount catalog validity unless operator policy allows shorter/overlap-safe windows only.	Avoid runtime surprises.
R-SIP-DA-05	Duplicate active assignment for same discount/scope/window is blocked unless duplicate policy allows stacking.	Prevent accidental double discount.
R-SIP-DA-06	Stacking group rules must prevent incompatible discounts from applying together.	Revenue protection.
R-SIP-DA-07	Manual high-value or long-duration assignments require approval.	Governance.
R-SIP-DA-08	Cancellation affects future runtime only; past invoice/wallet impact requires BIL adjustment.	Financial audit.
R-SIP-DA-09	Bulk assignment must be idempotent per target.	Prevent duplicate assignments on retry.
R-SIP-DA-10	Sales app can request or display only discounts allowed for the agent/franchise/territory.	Channel control.
R-SIP-DA-11	Approval policy and tasks must be handled through EM-CFG-04. SIP-03 only stores local approval reference and applies final outcome.	Shared approval governance.
R-SIP-DA-12	Camunda is used only when EM-CFG-04 policy mode is BPMN; simple single-step approvals may be DB-driven by EM-CFG-04 without Camunda.	Operational simplicity.

10. Rule Engine Guidance

SIP-03 should use Drools for operator-variable commercial decisions:

Rule file	Purpose
discount-assignment-eligibility.drl	Validates target eligibility against discount, package, franchise, channel, customer segment, and reason code.
discount-assignment-stacking.drl	Decides whether the new assignment conflicts with existing active assignments.
discount-assignment-approval.drl	Decides when to call EM-CFG-04 and which approval policy code to request.

Fixed checks such as non-null fields, date parsing, idempotency, and foreign id lookup remain normal application validation.

Example rule:

```
rule "Retention discount above 90 days requires approval"
when
    $a : DiscountAssignmentFact(reasonCode == "RETENTION_SAVE", durationDays > 90)
    $out : DiscountAssignmentDecision()
then
    $out.setApprovalRequired(true);
    $out.setApprovalPolicyCode("DISCOUNT_ASSIGNMENT_RETENTION");
end
```

11. Frontend Usage

11.1 FE-APP-01 Backoffice

Backoffice must support:

- discount assignment search by customer, subscription, campaign, status;
- create manual assignment with preview;
- approval queue;

- cancel/suspend assignment;
- bulk upload/batch monitoring;
- effective assignment view on customer 360 and subscription pages;
- runtime application drill-through link to DIS-OP/BIL when available.

11.2 FE-APP-02 Outdoor Sales / Franchise App

Sales app must support:

- show eligible acquisition discounts for lead/order/package;
- show discount terms in customer quote;
- submit campaign/order-bound assignment request;
- work offline with cached eligible discount summaries where allowed;
- sync assignment request when online.

The sales app must not invent discounts. It displays discounts returned by SIP-03/SIP-05 eligibility APIs.

12. Infra Recommendation

For a medium budget deployment, run SIP-03 inside the same commercial/catalog API deployment as SIP package management or the billing discount submodule, depending on final code ownership. Do not create a separate pod at first.

Component	Recommendation
API	Existing catalog/commercial API pods expose <code>/api/discount-assignments</code> .
DB	Same PostgreSQL cluster as catalog/billing commercial config.
Workers	Existing worker pool runs expiry scan and bulk assignment jobs.
Events	Kafka/MSK topics under <code>sophix.commercial.discount-assignment.*</code> .
Cache	Runtime read model/cache for active assignments by customer/subscription.

AWS deployment: use existing ECS/EKS service and RDS cluster, with separate schema if the team wants ownership clarity. Add CloudWatch alarms for failed bulk jobs, high duplicate rejection rate, expiry worker failure, and runtime query latency.

13. Testing Requirements

13.1 Unit Tests

- Discount catalog eligibility validation.
- Scope target validation.
- Duplicate active assignment prevention.
- Stacking group conflict.
- Approval policy decision.
- EM-CFG-04 approval evaluation/request/callback integration.
- Drools eligibility/stacking/approval decision.
- Cancellation does not reverse past application.

13.2 Integration Tests

- Create assignment after reading PLM-CFG-04 discount.
- Query effective assignments for DIS-OP.
- Bulk assignment creates per-target results.
- Cancel assignment and verify runtime query excludes it.

- Sales app eligibility query respects franchise/territory.

13.3 End To End Tests

- Retention agent creates approved subscription discount; next billing runtime sees it.
- Campaign creates order discounts in bulk; failed rows are reviewable.
- Cancelled discount stops future invoices but existing invoice remains unchanged.

14. Definition Of Done

- Discount assignment tables and APIs exist.
- Assignment validates against discount catalog and target owner modules.
- Approval and batch assignment flows work.
- EM-CFG-04 is used for configurable approval policies; SIP-03 does not hardcode approver steps.
- Drools rule files exist for configurable eligibility, stacking, and approval decisions.
- Runtime query API returns active assignments for DIS-OP.
- Events are emitted for assignment state changes.
- Backoffice can create, approve, cancel, and monitor assignments.
- Sales app can show/request eligible discounts.
- Tests cover manual, campaign, bulk, cancellation, and runtime query paths.